

MTE 204 Problem Set #1

Roots of Equations (Chapter 5, 6, 8 in Chapra 8th edition, International Version)

* For hand calculations, 3 iterations are sufficient. It is recommended to use MS Excel (or MatLab) for additional iterations to achieve convergence.

1. Graphical and Bisection Methods

Problems 5.1, 5.5, 5.14

2. Fixed Point Iteration, Newton-Raphson, Secant Methods

Problems 6.2 a – d), 6.4, 6.11, 6.19

3. Applications

Problems 8.4, 8.44